Lecture 08: Messaging Queue | Kafka | Rabbit MQ

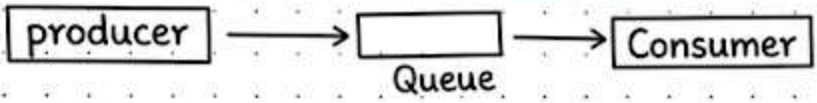
part - 1

Kafka - Messaging Queue

↳ It is not just a MQ

- # Messaging Queue

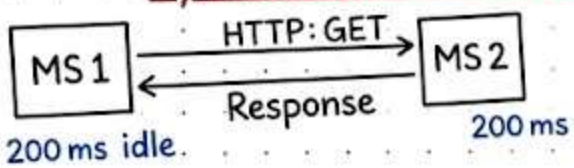
producer/consumer Architecture:



Benefits of producer/consumer

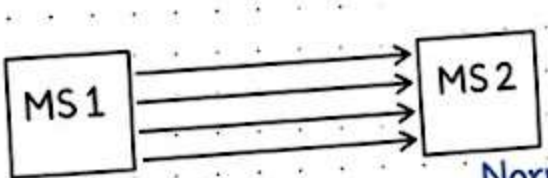
producer and consumer can work on there own pace.

Synchronous Communication



Async Communication





Busy service

High throughput

Normal service

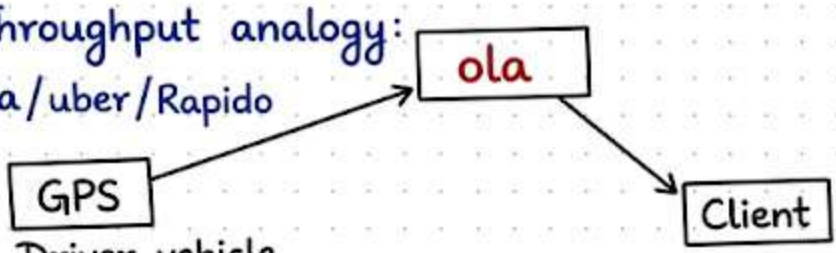
Throughput is less

MS 2 down

HLD → Throughput → OPS (operations per second)

- Throughput analogy:

ola/uber/Rapido



freq = 1 signal/sec

Back of the envelope:

Delhi → 1 lakh driver in ola
per sec → 1 lakh req/sec

Add value to Db:

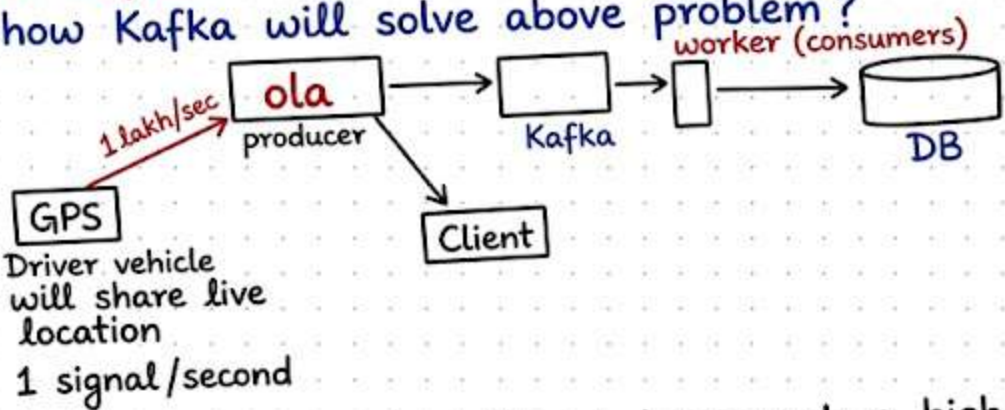
Fare calculation

1 Db will have very high load

Message Queues

1. Kafka

how Kafka will solve above problem?



- Every DB that exists will not have such a high throughput
- Kafka: Kafka is a MQ, that have a very high throughput
- workers will do bulk upload on DB

Job of a web server:

Take the request from client
process the request
put the data in DB

Dbs are comparatively slower.
latency will be higher

Why it is useful ?

1. pace matching



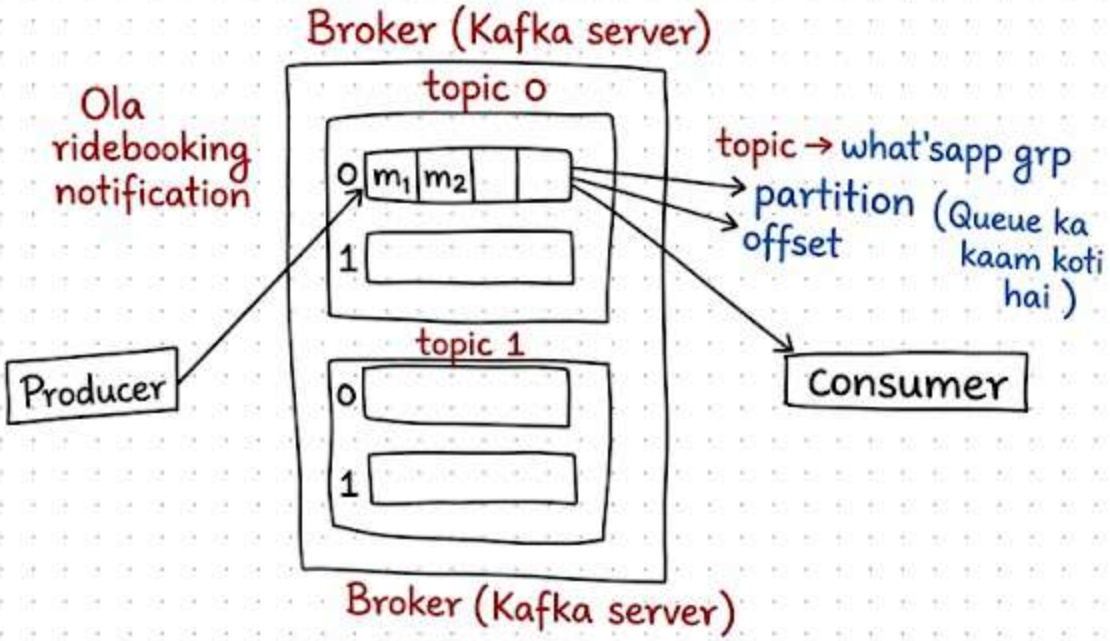
↳ Kafka
↳ It has high throughput

2. Async communication
   ↳ produce can do another work , after giving request

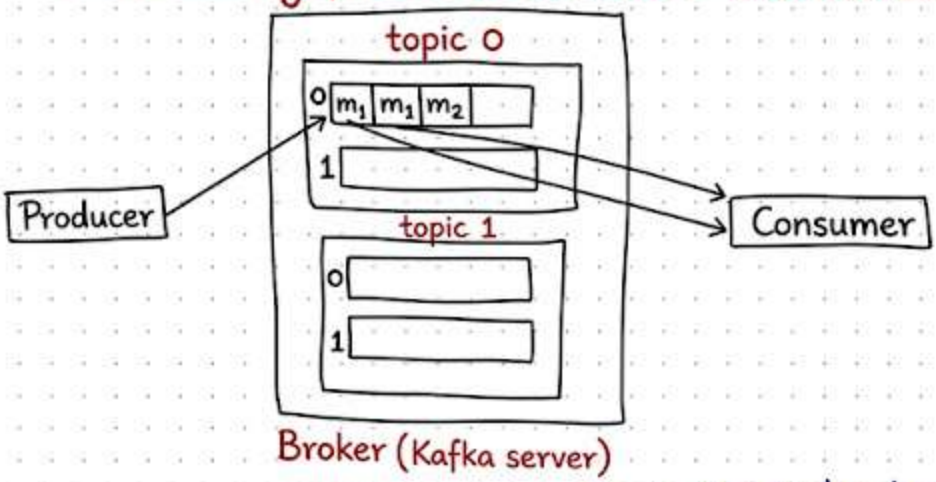
In depth architecture of Kafka

cluster
broker
Topic
Partition
offset
} Terms we will talk about

zookeeper, kraft
Consumer
consumer group
producer

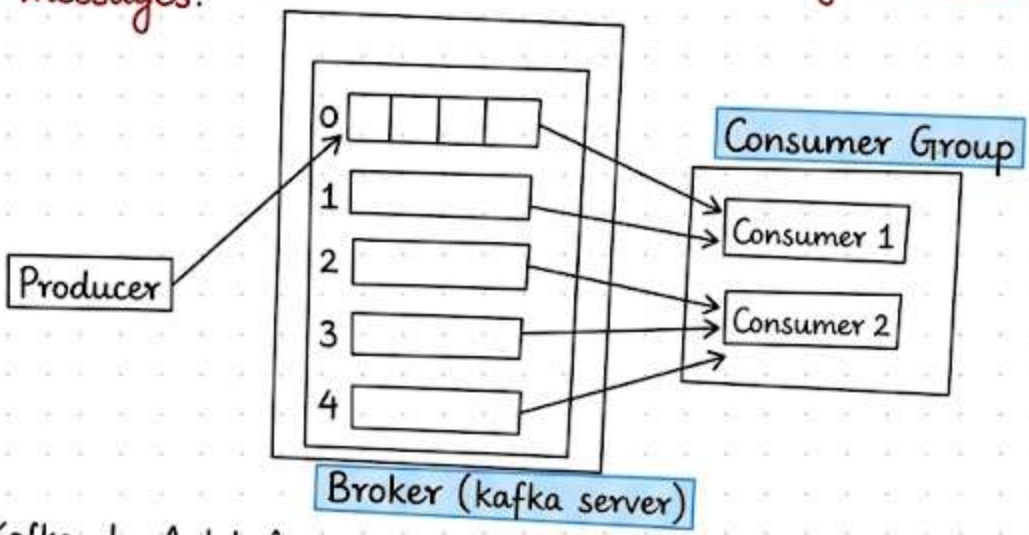


Introducing producer consumer architecture

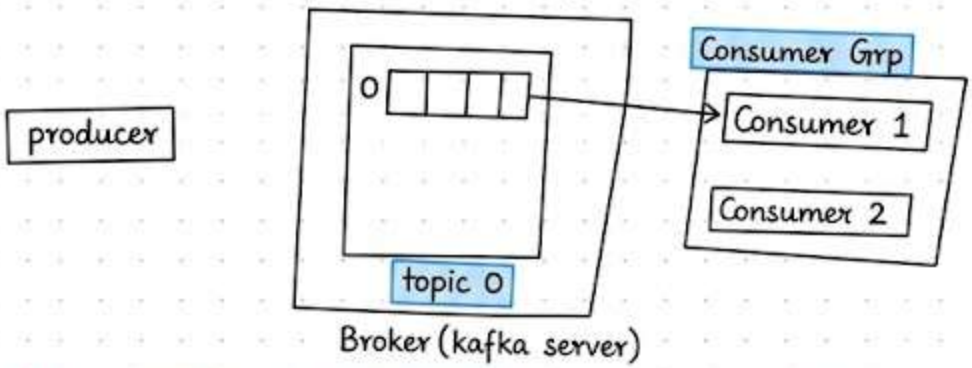


topic 0 → rider loc
topic 1 → notifications

Consumer Groups & How consumer actually consumes messages.

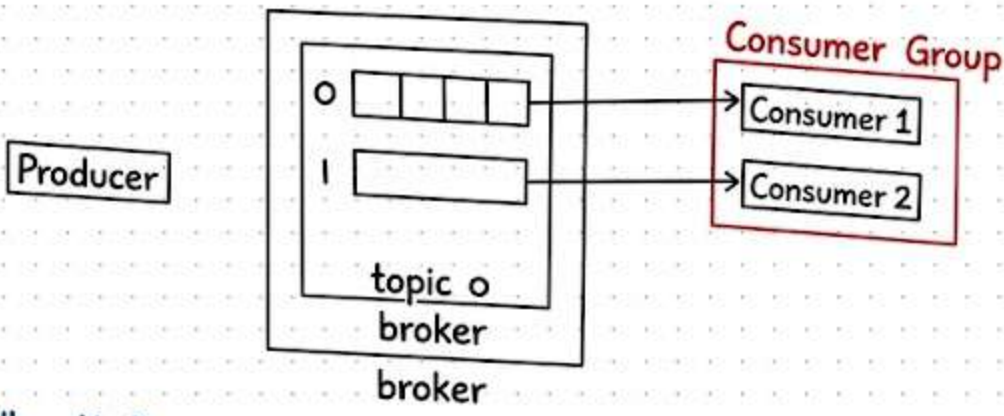


Kafka do Autobalancing, partition 4 can be casigned to any of 1 consumer, depends on broker



- if a number of partitions are less then number of consumers, some consumers will remain idle.
- 1 Consumer can consume multiple partitions. But, 1 partition can be consumed by only 1 Consumer.

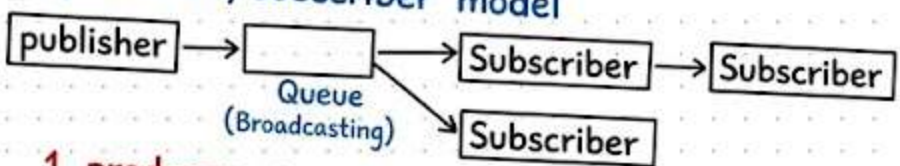
Consumer Group Architecture



Why Kafka does all this?

- Consumer Group is fairly new in Kafka
- Messaging queue serve 2 purpose:
  1. Queue (FIFO) → earlier:
  2. pub / sub
     ↳ publisher / subscriber model

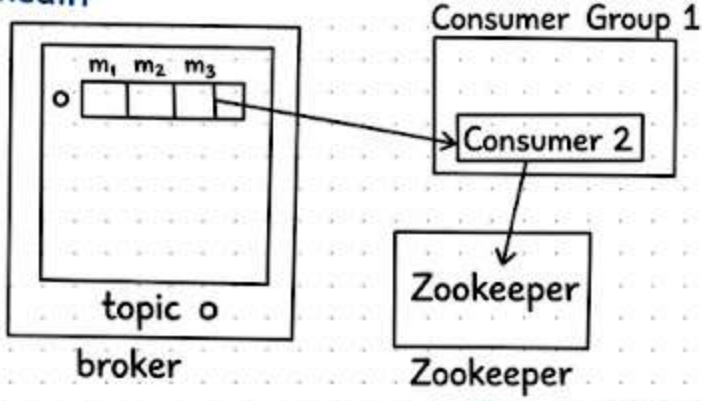




1 producer → many consumer

kafka → Linkedin

Zookeeper Integration

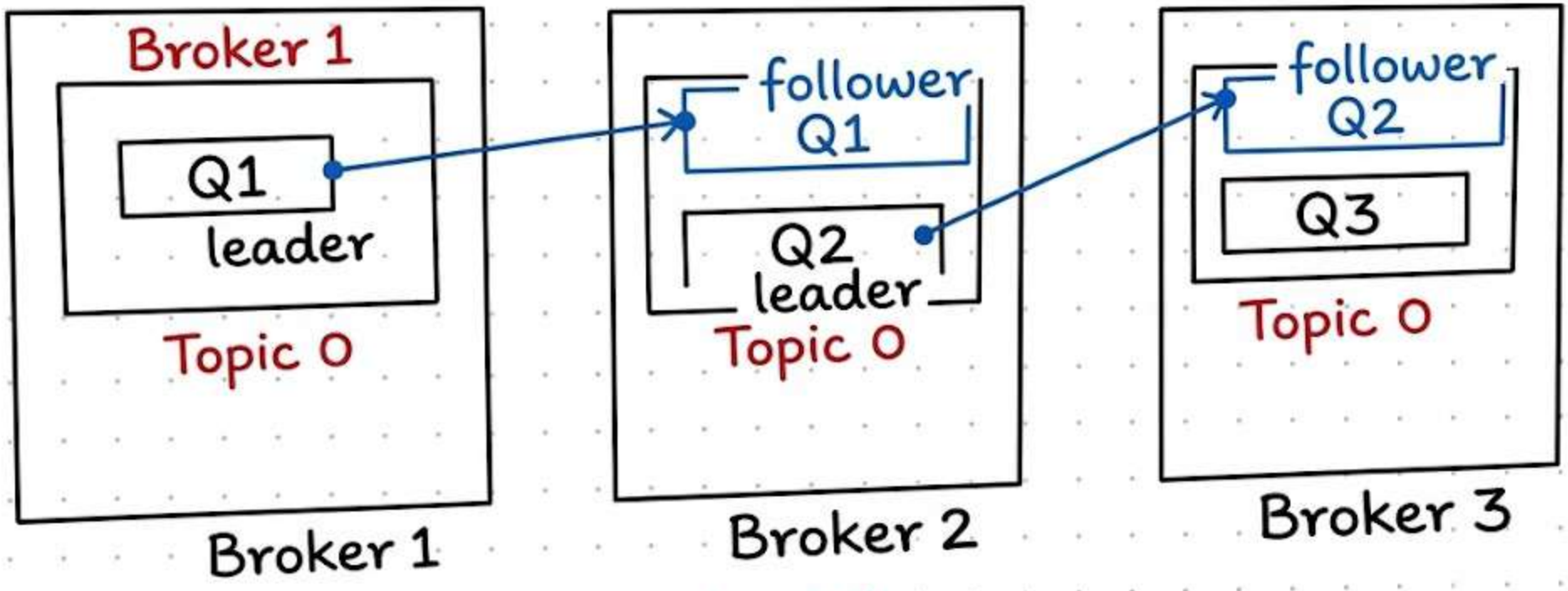


Consumer Group : 1
Consumer : 2
Topic : 0
Partition : 0
offset : 2

Clusters - Group of brokers

↳ Alag-Alag broker main split kar diya (Q_1,Q_3) mean partition kardiya - taki ek hi broker par load na aye.

↳ replicated - main info - leader;
copied info - follower



Leader → Both Read & Write
Follower → will just follow (listen) the leader